AI-BASED EMERGENCY CALL AND SMS TRIAGE SYSTEM FOR DISASTER RESPONSE IN SRI LANKAN LOW-RESOURCE LOCAL LANGUAGES

COMMON INTEGRATED SOLUTION REPORT (GROUP R26DS-011)

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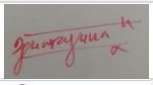
Sri Lanka

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

During floods, landslides and cyclones, Sri Lanka's emergency hotlines receive surges of calls and text messages in Sinhala, Tamil, English and code-mixed Singlish or Tanglish. Callers are frightened, omit critical details, and many people report the same incident, so operators must question, prioritise and de-duplicate reports by hand at the moment when time is shortest. Existing dispatch and crisis-informatics tools assume English, a single input channel and one record per call. This report presents an integrated triage system built by four students as four interdependent components. A multilingual conversational agent (C1) transcribes each caller turn, extracts incident entities and asks targeted clarification questions until the report is dispatch-ready. A multimodal urgency model (C2) fuses 91 acoustic features with transcript features to rate every call High, Medium or Low. A low-bandwidth SMS service (C3) cleans noisy messages with hallucination-guarded language models, asks adaptive follow-up questions and assigns a Medical Priority Dispatch System code. An incident-fusion service (C4) receives both streams through a versioned contract, reconciles the extracted fields, scores geographic, temporal, semantic and evidence agreement against open incidents, and makes a cost-sensitive merge decision that is shown, with its evidence, on a real-time operator dashboard. Components were evaluated on simulated corpora and on 59 real Disaster Management Centre calls. The urgency model reached 66.4% accuracy on simulated calls and 80.0% on scope-matched real calls; a language-model safety net cut dangerous SMS under-triage from 11 to 1 of 26 cases; and incident fusion made 0.9% false merges with 98.1% merge precision on held-out test reports. Recognition quality on real telephony audio and missed cross-lingual links remain the main limitations. The work shows that a human-in-the-loop, multi-channel triage pipeline for low-resource Sri Lankan languages is technically feasible and identifies the evidence needed before operational deployment.

Keywords: Emergency triage, Low-resource languages, Conversational AI, Speech urgency detection, Multi-source incident fusion

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We are grateful to the Disaster Management Centre of Sri Lanka for granting access to anonymised emergency call recordings from the November and December 2025 floods and landslides. These recordings made it possible to test the system on real calls rather than on simulated data alone.

We thank the volunteers who recorded simulated emergency calls and messages in Sinhala, Tamil and English, and the native speakers who reviewed question templates and transcripts. We also thank the academic and technical staff of the Sri Lanka Institute of Information Technology for the knowledge, facilities and computing resources that supported this research, and the members of the progress review panels whose feedback led to measurable improvements in every component.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
ASR	Automatic Speech Recognition
AUC	Area Under the Receiver Operating Characteristic Curve
B^3	B-cubed clustering precision, recall and F1
CAD	Computer-Aided Dispatch
CAP	Common Alerting Protocol (OASIS)
CPU	Central Processing Unit
CRF	Conditional Random Field
CV	Cross-Validation
DEV	Development (validation) data split
DMC	Disaster Management Centre, Sri Lanka
EOC	Emergency Operations Centre
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LaBSE	Language-agnostic BERT Sentence Embedding
LLM	Large Language Model
LR	Logistic Regression
mDeBERTa	Multilingual Decoding-enhanced BERT with Disentangled Attention
MFCC	Mel-Frequency Cepstral Coefficient

MPDS	Medical Priority Dispatch System
mT5	Multilingual Text-to-Text Transfer Transformer
NER	Named Entity Recognition

Abbreviation	Meaning
NLI	Natural Language Inference
NLP	Natural Language Processing
REST	Representational State Transfer
RF	Random Forest
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SMS	Short Message Service
STT	Speech-to-Text
TTS	Text-to-Speech
XLM-R	XLM-RoBERTa cross-lingual language model

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Sri Lanka is exposed to recurrent natural hazards. The south-west and north-east monsoons bring floods and landslides every year, cyclones and storm surges affect the coast, and several hazards often occur together. When a large event strikes, the population turns to emergency hotlines. The Disaster Management Centre (DMC), established under the Sri Lanka Disaster Management Act No. 13 of 2005 [1], coordinates the national response and operates an Emergency Operations Centre (EOC) that receives calls from the public. The Sendai Framework for Disaster Risk Reduction 2015–2030 identifies access to timely disaster risk information as a priority for reducing loss of life [2]; how quickly emergency reports are understood and acted upon is a direct part of that capability.

Emergency communication in Sri Lanka has characteristics that make it difficult to automate. Callers speak Sinhala, Tamil or English and frequently mix them, either in speech or in Romanised text (Singlish and Tanglish). They are often panicking, speak over background noise on narrow-band telephone lines, and describe their location with landmarks and local names rather than addresses. When voice networks are congested or batteries are low, people send short SMS messages instead, written with abbreviations, spelling variation and emoji. Finally, a single incident such as a flooded road may be reported by many people within minutes, each in a different language and with a different description of the same place.

The current practice at an EOC is largely manual and first-come, first-served. A trained operator answers each call, asks follow-up questions, judges urgency, writes a record and passes it on for dispatch. This works in ordinary conditions, but during a surge the same operators must also recognise which reports describe the same event, decide which callers are in the most danger, and keep track of incomplete reports. Commercial Computer-Aided Dispatch (CAD) systems used in high-income countries treat each call as a separate record, assume a single dominant language and a geocoded address database, and do not link multiple reports of one event [3]. Research systems for crisis informatics, such as AIDR [4], HumAID [5] and CrisisTransformers [6], process

public social-media posts, mostly in English, rather than private calls and SMS sent to authorities [7].

Sinhala and Tamil are low-resource languages for speech and language technology. Multilingual models such as Whisper [8], XLM-RoBERTa [9] and LaBSE [10] include them, but published work shows that such models perform markedly worse on languages with little training data [11], and no published emergency-domain dataset or system exists for Sinhala or Tamil calls. Recent advances in speech recognition, multilingual transformers and large language models (LLMs) nevertheless make it feasible to build a decision-support system that helps operators clarify, prioritise and consolidate multilingual emergency reports. This project, R26DS-011, designs, builds and evaluates such a system.

1.2 Stakeholders and user needs

The stakeholders of the integrated system were identified during the proposal stage and refined through the progress reviews and the data-access process with the DMC. Table 1.1 summarises them and the needs that the solution addresses.

Table 1.1: Stakeholders of the integrated triage system and their needs

Stakeholder	Role	Principal needs
Callers and SMS senders (disaster victims)	Primary	Be understood in their own language; be asked few, simple questions; have a life-threatening situation recognised quickly; be able to report by SMS when calls fail
DMC call-takers	Primary	Complete reports with location, people, injuries and help needed; less repetitive questioning; urgency cues they can see and override
EOC dispatchers and supervisors	Primary	One record per real incident instead of many duplicates; a ranked, explainable incident picture; uncertain decisions flagged for review
First responders (fire, police, ambulance, relief teams)	Secondary	Accurate, consolidated incident details and locations before deployment
DMC informationtechnology staff	Secondary	A system that runs on DMC-controlled infrastructure, degrades gracefully and can be audited
Tamil-speaking and rural communities	Secondary	Equal quality of service regardless of language or region

SLIIT supervisors and the research community	Secondary	Reproducible methods, honest baselines and documented limitations

Engagement with the DMC confirmed two needs that shaped the design. First, the DMC provided 60 anonymised recordings of real emergency calls made during the November–December 2025 floods and landslides (59 usable after one corrupted file was excluded). Listening to these calls showed how often callers give incomplete information, how much telephony audio differs from simulated recordings, and that operators must ask the same questions repeatedly. Second, the progress review panels required that every automated decision be explainable and that the operator remain the final decision-maker. The resulting user needs are: understand callers in Sinhala, Tamil, English and code-mixed forms; collect missing critical facts with as few questions as possible; surface urgency from both what is said and how it is said; accept reports over SMS; consolidate duplicate reports across channels and languages without merging separate emergencies; and show every decision with its evidence.

1.3 Alignment with the Sustainable Development Goals

The project is aligned with the United Nations 2030 Agenda for Sustainable Development [12] through the goals and targets listed in Table 1.2. The contribution claimed is to the capability of emergency services to handle reports during disasters; the project has not yet measured impact on deaths or losses, which would require operational deployment.

Table 1.2: Sustainable Development Goal targets addressed by the project

SDG target	Intended contribution
SDG 11, Target 11.5: reduce deaths, people affected and economic losses caused by disasters	Faster clarification, prioritisation and consolidation of emergency reports during disasters
SDG 13, Target 13.1: strengthen resilience and adaptive capacity to climate-related hazards	Decision support for monsoon floods and landslides, the hazards that dominate the DMC call data
SDG 3, Target 3.d: strengthen capacity for early warning, risk reduction and management of health risks	Recognition of life-threatening situations (entrapment, injury, breathing difficulty) for earlier response
SDG 10, Target 10.2: promote the social inclusion of all, irrespective of language or origin	Equal access for Sinhala, Tamil and code-mixed speakers, with language gaps reported openly

1.4 Purpose and scope of the common report

This common report presents the shared problem, the integrated solution, the way the four components operate together, and the evaluation of the solution as a whole. Each component is summarised in sufficient detail to show its role and its principal results, but the detailed literature review, methodology, experiments and negative results of each component are reported in the four individual theses of the team members listed on the title page.

The report is organised as follows. Chapter 2 defines the overall problem, its root causes and the four subproblems. Chapter 3 describes the solution architecture and summarises each component. Chapter 4 explains how the components are integrated, including interfaces, shared non-functional requirements and exception handling. Chapter 5 describes implementation and project execution. Chapter 6 reports integrated testing, results and validation against the group objectives. Chapter 7 discusses commercialisation, sustainability and responsible innovation, and Chapter 8 concludes with future directions.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem addressed by the team is stated as follows:

During disaster surges, Sri Lankan emergency services receive large volumes of calls and SMS messages in Sinhala, Tamil, English and code-mixed forms that are incomplete, differ in urgency in ways that are not recorded, and frequently describe the same incident many times. Because these reports are clarified, prioritised and consolidated manually and one at a time, critical information is collected slowly, the most urgent cases are not reliably identified, and duplicate records obscure the true incident picture at the time when response capacity is most constrained.

The problem is defined independently of any particular technology. It concerns four observable deficiencies in how reports reach a dispatcher: missing facts, unrecorded urgency, an unusable channel when voice fails, and fragmentation of one event into many records. The integrated research question that follows is: *how can an AIsupported triage system clarify, prioritise and fuse multilingual emergency calls and SMS messages in Sri Lankan low-resource languages, while keeping every decision explainable and under human control?*

2.2 Root causes and overall research gap

Table 2.1 summarises the root causes identified from the literature, the DMC recordings and the simulated data, together with the limitation of current practice for each.

Table 2.1: Root causes of the common problem and limitations of current practice

Root cause	Evidence	Limitation of current practice
Callers omit critical facts under stress	Most simulated and real calls lack at least one of location, people count, injury or entrapment at the first turn	Operators ask fixed questions one by one; CAD forms only record what is said
Urgency is carried by voice as well as words	Calm callers can describe critical situations, and distressed callers may give little content	Records capture text only; prosody is lost after the call
Language and script	Sinhala, Tamil, English, Singlish,	Dispatch and crisis-NLP tools

Root cause	Evidence	Limitation of current practice
diversity	Tanglish and mixed-script messages in both channels	assume English and standard spelling [3], [7]
Voice-network congestion during disasters	SMS is used when calls fail; messages are short, noisy and abbreviated	SMS reports are handled as free text without structured extraction or follow-up
Many reports of one event	Several callers describe the same place in different languages and scripts	Each call becomes a separate record; operators de-duplicate by memory
Low-resource language technology	General multilingual models are weaker for Sinhala and Tamil [11]	No emergency-domain datasets or models exist for these languages

The overall research gap follows from these causes. Previous work addresses parts of the problem in isolation and almost exclusively for high-resource languages: speech emotion and urgency models for English or Mandarin, dialogue systems with fixed slot-filling, crisis-message classification for English social media, and incident clustering on geotagged posts using location alone. No published system combines adaptive multilingual clarification of calls, multimodal urgency estimation, lowbandwidth SMS triage and cross-lingual, cross-channel incident fusion for Sinhala and Tamil, and no published work evaluates such a pipeline on real Sri Lankan emergency calls. Because each deficiency degrades the others (an unclarified report cannot be fused reliably, and an urgency label is of little use if the same incident appears ten times), a four-component solution that shares one data contract is required.

2.3 Subproblem and component mapping

The common problem was decomposed into four non-overlapping subproblems, each assigned to one team member, as shown in Figure 2.1 and Table 2.2. The subproblems are interdependent: C2 analyses the caller turns that C1 captures, C1 and C3 produce the structured reports that C4 consolidates, and C4 presents the urgency outputs of C2 and C3 to the operator.

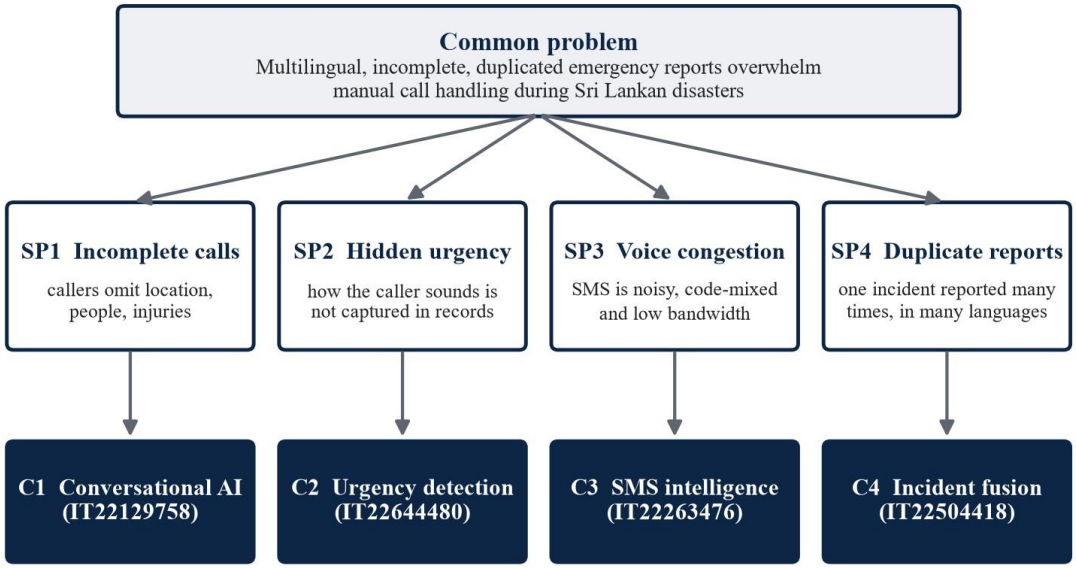


Figure 2.1: Decomposition of the common problem into four subproblems and components Table 2.2: Subproblem-to-component mapping

Subproblem	Component and owner	Why it is necessary for the common problem
SP1: emergency calls are incomplete and callers cannot be questioned efficiently in their own language	C1 Multilingual Emergency Conversational AI (IT22129758)	Without clarification, downstream urgency and fusion operate on partial facts
SP2: urgency conveyed by the caller's voice and words is not captured	C2 Multimodal Urgency Detection (IT22644480)	Operators need a consistent urgency signal for every call to decide whom to help first
SP3: SMS reports sent when voice fails are noisy and unstructured	C3 Low-Bandwidth SMS Emergency Intelligence (IT22263476)	Provides a second intake channel with structured fields and a dispatch priority
SP4: one incident becomes many records across channels and languages	C4 Multi-Source Incident Fusion and Operator Dashboard (IT22504418)	Turns report streams into one record per incident and shows the combined picture to operators

2.4 Overall aim and group objectives

Overall aim. To design, implement and evaluate an integrated, human-in-the-loop triage system that clarifies, prioritises and fuses multilingual emergency calls and SMS messages in Sri Lankan low-resource languages into explainable incident records for disaster-response operators.

The group objectives in Table 2.3 refine this aim. Each objective is measurable and mapped to the components that deliver it and to the evidence used in Chapter 6.

Table 2.3: Group objectives, responsible components and evaluation evidence

ID	Group objective	Components	Evaluation evidence
GO1	Accept emergency reports in Sinhala, Tamil, English and code-mixed forms on both the voice and the SMS channel, end to end	C1, C3, C4	Live call and SMS sessions processed through all components; per-language results
GO2	Reduce incomplete reports by asking targeted clarification questions in the caller's language	C1, C3	Question diversity and safety pass rate; readiness assessment; distress-detection benchmark
GO3	Attach an urgency or priority signal to every report, favouring recall of the most urgent cases	C2, C3	Accuracy, macro-F1 and Highclass recall on simulated and real DMC calls; MPDS under-triage
GO4	Consolidate duplicate reports across channels and languages with a falsemerge rate below 2% and merge precision above 90%	C4	False merge, false split, link recall, merge precision and B^3 F1 on held-out test reports
GO5	Make every automated decision explainable and leave final decisions to operators	C1–C4	Evidence shown for urgency, MPDS codes and merges; review flags; dashboard
GO6	Operate as one integrated system in which a slow or failed component never blocks intake	C1–C4	Versioned contract, live integration check, dropped-call and outage tests

2.5 Scope assumptions and constraints

In scope. Voice calls through a Twilio telephone number or a browser microphone; SMS through a Twilio number or a web chat; the incident types that dominate disaster calls (flood, fire, landslide, medical and related rescue needs); urgency on a High / Medium / Low scale for calls and the MPDS ECHO–OMEGA scale for SMS; incident consolidation and an operator dashboard; and a single-site research prototype running on a laptop.

Out of scope. Automatic dispatch of resources, integration with the DMC's internal systems, social-media monitoring, mobile applications for victims, and production

hosting, authentication and high-availability infrastructure. These are discussed as future work.

Assumptions. Callers and senders consent to their reports being processed for emergency response; real recordings are used only for evaluation under the DMC's permission; and the DMC operator remains responsible for every dispatch decision.

Constraints. Real emergency data are scarce and sensitive, so components were developed on simulated or synthetic corpora and tested on real data only where it was available. Computation was limited to a CPU laptop for inference and the free Google Colab GPU tier for fine-tuning. Some commercial fine-tuning platforms were unavailable to the team. External services (OpenAI, Google Places, Twilio) were used where justified, but every component was designed to keep working when they are switched off.

Dependencies. C2 depends on the caller audio and transcript of each C1 turn; C4 depends on the turn history and extracted fields published by C1 and C3; and the dashboard depends on C4's incident store.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The solution is a single triage pipeline with two intake channels and one consolidation and presentation layer. On the voice channel, C1 conducts the conversation with the caller and C2 runs inside it, rating the urgency of every caller turn and of the call as a whole. On the SMS channel, C3 conducts a text conversation with the sender and assigns a dispatch priority. Both channels publish each finished or abandoned conversation, together with their extracted fields and urgency signals, to C4. C4 decides whether each new report belongs to an incident that is already open or starts a new one, maintains a unified incident record, and presents incidents on a live dashboard.

The central value proposition is that the operator receives one explainable, prioritised record per real incident, built from all calls and messages about it, instead of a stream of incomplete, unranked and duplicated reports. Three design principles are shared by all components. First, the caller's own words are preserved: values are extracted from them, but never silently corrected, translated or invented. Second, the cost of errors is asymmetric: under-triage and merging two real emergencies are treated as more dangerous than over-triage and leaving a duplicate unmerged. Third, the human stays in control: uncertain cases are flagged for review and every decision carries its evidence.

3.2 High level architecture

Figure 3.1 shows the high-level architecture. The users are callers, SMS senders and DMC operators. C1 and C3 are separate Flask services with their own telephony webhooks; C2 is loaded as a Python module inside C1 so that urgency is computed on the same audio without a network hop; and C4 is a Flask REST service with a background worker, a report store, an incident store and two web views. A shared client module, used by both C1 and C3, publishes reports to C4 using a versioned contract. A launcher page opens all user-facing parts of the system.

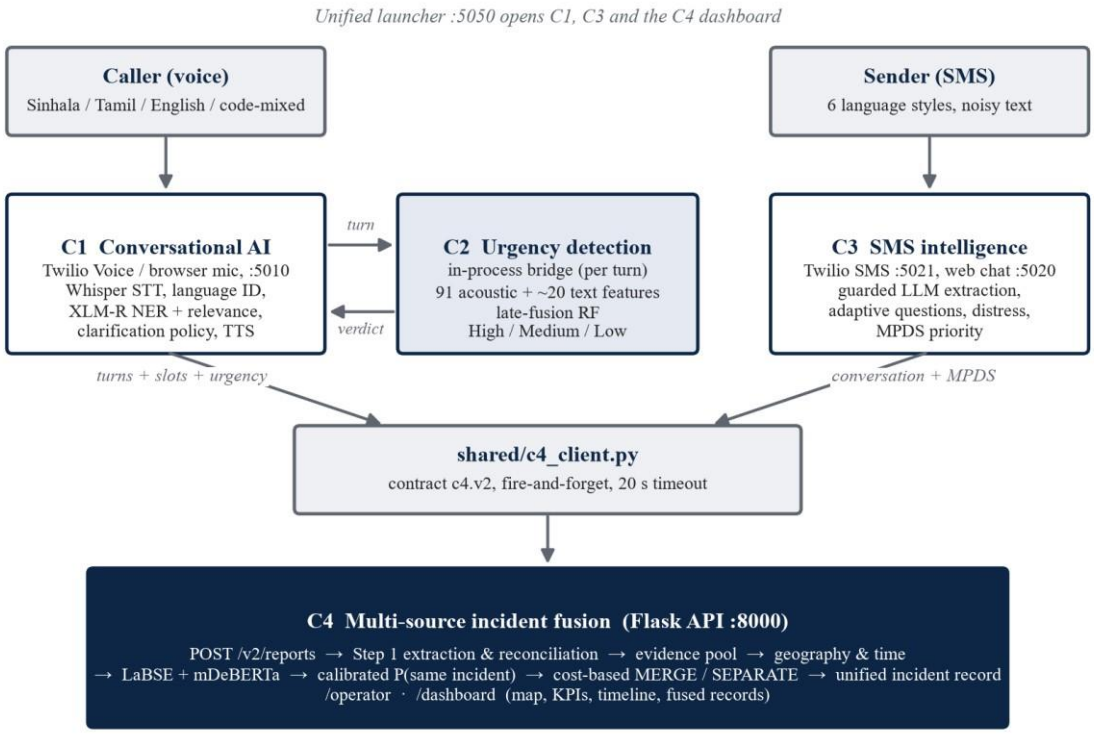


Figure 3.1: High-level architecture of the integrated triage system

The external systems are Twilio (voice and SMS gateway), OpenAI models (optional extraction, distress and severity checks, and the teacher model used to train offline students), Google Places (optional location resolution, cache-first) and OpenStreetMap tiles for the dashboard map. In the prototype, all services run on one Windows laptop and only the Twilio webhooks are exposed through a temporary tunnel during testing. The intended deployment boundary is an on-premises server controlled by the DMC, with a GPU for speech recognition.

3.3 Component 1 summary

Table 3.1 summarises the owner, subproblem, input, core method, output and contribution of Component 1.

Table 3.1: Summary of Component 1: Multilingual Emergency Conversational AI

Aspect	Description
Owner	Edirisinghe A.K. (IT22129758)
Subproblem	SP1: incomplete calls and inefficient multilingual questioning
Input	Caller audio per turn (Twilio Voice or browser microphone)

Core method	faster-whisper speech-to-text with a Sinhala-fine-tuned Whisper-small
Aspect	**Description**
	model and an STT guard; code-mix language identification; fine-tuned XLM-R named entity recognition and utterance-relevance models; multilingual numeral normalisation; protocol- and confidence-aware dispatch-readiness assessment; adaptive multi-agent clarification policy; language-aware text-to-speech
Output	The next spoken question in the caller's language; a final emergency JSON with slots, turn history and C2's urgency verdict, published to C4
Contribution	Turns a free-form, incomplete call into a dispatch-ready structured report and supplies the caller turns that C2 and C4 analyse

For each caller turn, C1 transcribes the audio, identifies the language among six codes (English, Sinhala, Tamil, Singlish, Tanglish and Sinhala–Tamil mixed), filters off-topic and non-informative utterances, extracts seven entity types (incident, area address, exact address, people count, trapped, place type and injury) and updates the dialogue state. A readiness assessor aligned with call-taking practice (where, what, who, when and life safety) and with OASIS CAP urgency, severity and certainty dimensions [13] decides which missing or low-confidence field matters most. The clarification policy then chooses among asking directly, asking softly, asking for refinement or asking nothing, and the question is spoken back with a voice that matches the caller's language. Safety nets finalise the report when the caller stops responding.

3.4 Component 2 summary

Table 3.2 summarises the owner, subproblem, input, core method, output and contribution of Component 2.

Table 3.2: Summary of Component 2: Multimodal Urgency Detection

Aspect	Description
Owner	Thanseel S.M. (IT22644480)
Subproblem	SP2: urgency conveyed by voice and words is not captured
Input	Audio of each confirmed caller turn and the transcript produced by C1
Core method	Noise reduction, 16 kHz resampling and silence trimming; 91 acoustic features (MFCC statistics, pitch, jitter, energy dynamics, zero-crossing rate, spectral centroid, speaking rate); about 20 linguistic and context features (keyword and context scores, repetition, ASR quality); two Random Forest models with SMOTE; late fusion of their probabilities (0.5 / 0.5)

Output	Per-turn and call-level urgency (High / Medium / Low) with confidence,
Aspect	**Description**
	class probabilities and peak severity
Contribution	Gives the operator and the incident record a consistent urgency signal for every call, including calls that are cut off

C2 was developed on 267 simulated calls labelled by two annotators (Cohen's kappa 0.695) and evaluated externally on 59 real DMC calls that were annotated before any model was applied and never used for training. Speaker diarisation with pyannote.audio [14] isolates the caller in the DMC recordings. The model was frozen before the DMC evaluation so that real-data results are untuned. Late fusion of an acoustic-only and a text-only model was chosen after comparison with early fusion and seven other fusion strategies; it is the only candidate that improved on all three evaluation views. In the live system, C2 runs inside C1 on every confirmed turn, and its call-level verdict combines the mean probability with the peak severity so that a single highly urgent turn is not averaged away.

3.5 Component 3 summary

Table 3.3 summarises the owner, subproblem, input, core method, output and contribution of Component 3.

Table 3.3: Summary of Component 3: Low-Bandwidth SMS Emergency Intelligence

Aspect	Description
Owner	Karunarathna D.M.S.G. (IT22263476)
Subproblem	SP3: noisy, unstructured SMS reports when voice fails
Input	SMS messages (Twilio) or web-chat messages in six language styles
Core method	Adaptive router that selects only the cleaning stages a message needs; LLM spelling, enhancement and grammar repair with hallucination guards; quote-verified field extraction; deterministic memory, answer parsing and multilingual question templates; LLM distress detection; fatigue-adaptive numbered questions; MPDS-based priority with an LLM safety net that can only raise urgency; an offline fine-tuned mT5-small model
Output	Incident type, location, people affected, help needed, an MPDS code (ECHO to OMEGA) with its evidence, a sorted dispatch queue, and the full conversation published to C4

Contribution	Provides a second, low-bandwidth intake channel that produces structured, prioritised reports comparable with voice reports

C3 was built on 864 SMS messages: 714 raw distress messages and 150 synthetic messages that cover English, Sinhala, Tamil, Singlish, Tanglish and mixed script evenly. A noise taxonomy derived by comparing successive cleaning stages showed that broken grammar affects 83.7% of messages, more than spelling errors (37.2%) or code-mixing (40.4%). Language models are used only where rules were shown to fail (extraction, distress and severity), and every model output must cite a verbatim quote from the message that is checked before the value is accepted [15]. Priority follows the conceptual structure of the Medical Priority Dispatch System, which is designed for remote triage from a caller's description [16]. A watchdog publishes partial cases when the sender stops replying.

3.6 Component 4 summary

Table 3.4 summarises the owner, subproblem, input, core method, output and contribution of Component 4.

Table 3.4: Summary of Component 4: Multi-Source Incident Fusion and Operator Dashboard

Aspect	Description
Owner	Siriwardena A.N. (IT22504418)
Subproblem	SP4: one incident becomes many records across channels and languages
Input	c4.v2 reports from C1 and C3: ordered turns, native upstream fields, C2 urgency or C3 MPDS code, and status flags
Core method	Input adapter; question-anchored answer recovery; mT5-small student extraction (trained on guard-accepted GPT-4o-mini outputs); three-source reconciliation; verbatim evidence pool categorised by a learned reader; location resolution and calibrated geographic scoring; incident-typespecific temporal scoring; fine-tuned LaBSE similarity; mDeBERTa evidence judge; gradient-boosting fusion to a calibrated P(same incident); cost-sensitive MERGE / SEPARATE decision at incident level; ambiguity review flag
Output	Unified incident records with consolidated fields, count ranges, timelines and evidence trails; an operator dashboard with map, KPIs, timeline and per-incident drill-down
Contribution	Converts the outputs of C1, C2 and C3 into one explainable, prioritised record per real incident

C4 was developed and evaluated on SL-DisasterCoref, a benchmark built by the team of 535 multilingual call and SMS reports grouped into gold incidents and split by incident into 308 training, 112 development (DEV) and 115 test reports. It contains 1,445 question–answer evidence items and 1,156 labelled evidence pairs, with hard negatives such as adjacent houses on the same road and the same place at a different time. Every setting was chosen on DEV and the test split was used once per decision. Two rules declared by the researcher constrain the design: urgency and priority never enter a merge decision (they are shown to the operator only, and a test enforces this), and a false merge is treated as three times as costly as a false split.

3.7 End to end user and data flow

Figure 3.2 traces a report through the system. On the voice channel, C1 asks an opening question, transcribes each caller turn, extracts entities, obtains C2's urgency for the turn and asks for the most important missing fact, repeating until the report is dispatch-ready, the caller ends the call or the call drops. On the SMS channel, C3 cleans and extracts the first message, checks each reply for distress or fatigue, and sends follow-up questions until the case is complete or abandoned. Both channels then publish to C4.

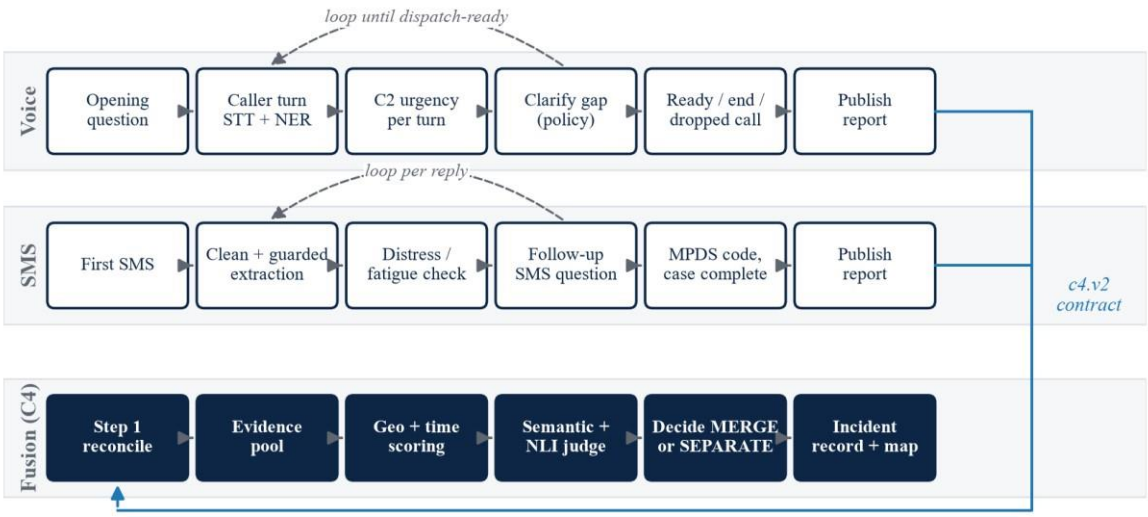


Figure 3.2: End-to-end flow of a call and an SMS report through the integrated system

A concrete example from the live integration test illustrates the interaction. An SMS in Sinhala script reporting an emergency at “14 පන්සල පාර ගම්පල” (14 Pansala

Road, Gampaha) opens a new incident. A second report written in Romanised Singlish, “14 pansala para gampaha”, is scored against the open incident with P(same incident) of 0.93 and is merged, although the two reports share no characters. A third report about “23 පන්සල පාර”, the same road but a different house, receives P = 0.006 and opens its own incident. The dashboard shows two incidents, one of them with two fused reports, the evidence that supported the merge, and the highest urgency reported by any member report.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

The team followed an interface-first integration strategy. The components were developed in parallel by their owners, but the data each component publishes and consumes was agreed early and frozen in a versioned contract. Three integration decisions follow from the architecture.

- **In-process urgency.** C2 is imported by C1 through a bridge module rather than called over HTTP. Urgency is computed on exactly the audio and transcript that C1 confirmed for a turn, models are pre-loaded with a warm-up call, and an asynchronous entry point keeps the conversation responsive.
- **One shared publishing client.** Both intake channels use the same module, shared/c4_client.py, so the mapping from each component's native data to the C4 contract exists in one place only. Each component forwards its own field block under its native field names; C4 maps them and never corrects them.
- **Fire-and-forget publishing.** Reports are sent on a daemon thread with a 20second timeout, and all exceptions are caught. A slow or unavailable C4 therefore cannot delay or break a live call or SMS conversation.

The contract evolved once. Version 1 sent one composed text string per report. When C4 adopted evidence-based matching, which compares operator questions paired with caller answers, version 2 (c4.v2) replaced the string with the ordered conversation turns. The change was made in the shared client and in C4 together, the version 1 client was retained as a backup, and C3 was extended to keep an untruncated conversation log because its fatigue agent kept only the last five turns. Integration risks were managed by a strict schema that rejects unexpected fields, an automated live integration check that starts a real C4 server and publishes C1 and C3 sessions through the real client, and offline bridge and wiring tests for the C1–C2 connection.

4.2 Interfaces data contracts and shared models

Table 4.1 lists the interfaces between the components and the error-handling responsibility for each. Figure 4.1 shows the corresponding message sequence. **Table 4.1: Interfaces between components**

Interface	Mechanism	Payload	Error handling

C1 → C2 (per turn)	Python call, async: classify_urgency_late_fusion_async()	Turn audio and C1 transcript	Failure leaves the turn without urgency; the call continues
C1 → C2 (end of call)	end_call(), also triggered by idle timeout	Accumulated turn results	Verdict computed for dropped and abandoned calls
C1 → C4	shared client, POST /v2/reports	Turn history, slots, C2 verdict	Fire-and-forget, 20 s timeout, exceptions swallowed
C3 → C4	shared client, POST /v2/reports	Conversation log, memory fields, MPDS code	As above; partial cases published by a watchdog and later superseded
Dashboard → C4	GET /v2/incidents, GET /v2/reports (polled every 5 s)	Incidents, links, decisions, fused records	Stale view until the next successful poll
Twilio → C1 / C3	Voice and SMS webhooks	Audio stream or SMS body	Twilio retries; components return valid TwiML

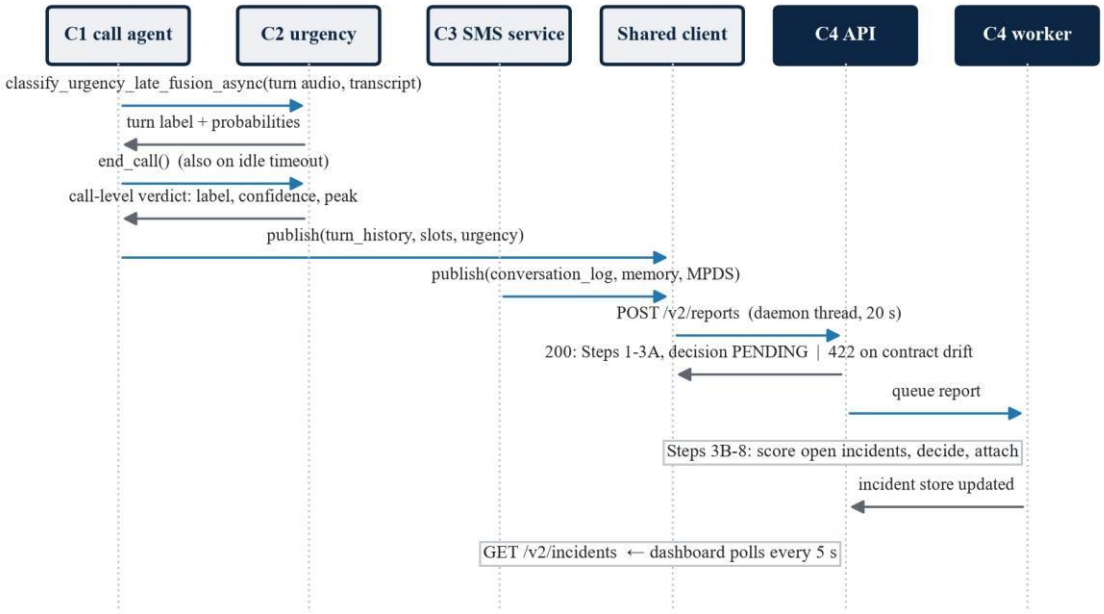


Figure 4.1: Message sequence between the components

The c4.v2 contract is defined as a Pydantic model in C4. Its fields are listed in Table 4.2; Appendix A gives an illustrative payload. The report and each turn reject unknown fields, so contract drift fails loudly with HTTP 422 instead of being silently ignored. The urgency verdict ignores extra diagnostic fields, so C2 can add diagnostics without

breaking C4. Language codes reported by C1 (for example siLK, ta-IN, si-en) are normalised by the shared client to plain names (sinhala, tamil, singlish).

Table 4.2: Fields of the c4.v2 intake contract

Field	Type	Meaning
contract_version	"c4.v2"	Version tag; any other value is rejected
report_id	string	Identifier assigned by the publishing component
source_component, channel	enumerations	Publishing component and channel (call or sms)
arrival_time	datetime, optional	Omitted by default; C4 stamps its own receipt time
language	string, optional	Normalised language name
turns	list of {speaker, text, turn_no, event}	Ordered conversation; speaker is caller or system; event marks, for example, a silence timeout
upstream	object	The publishing component's extracted fields under their native names
urgency	object, optional	C2 verdict: label, confidence, probabilities, peak label, number of turns
priority_code	string, optional	C3 MPDS code, ECHO to OMEGA
disconnected	boolean	Call cut or conversation abandoned
case_status, supersedes	strings, optional	C3 partial or complete case, and the partial case that a complete case replaces

Shared models and identifiers are kept deliberately small. Each report carries its own identifier; C4 assigns incident identifiers and stores links between reports and incidents. Dataset labels are held only in an evaluation metadata block that no processing stage may read, which prevents label leakage. The structured fields of C1 and C3 differ in name and granularity, and C4 maps both to one internal report model with explicit missing values: a missing count is recorded as unknown, never as zero.

4.3 Shared nonfunctional requirements

Table 4.3 summarises how the shared non-functional requirements are addressed across the components and where the prototype falls short.

Table 4.3: Shared non-functional requirements and how they are met

Requirement	How it is addressed	Current status
Performance	C4 replies with extraction and location results within the 20 s client budget (about 4 s in tests) and runs heavier matching in a background worker (about 0.3 s per aligned evidence pair). Models are pre-loaded at startup.	C1 speech recognition takes about 50–60 s per turn on a CPU laptop; a GPU or streaming STT is required for live use
Reliability	Fire-and-forget publishing; idle-finalise (120 s) and auto-commit (90 s) safety nets for dropped calls; SMS abandonment watchdog; incident store survives restarts	Met in the prototype; no redundancy or failover
Security	Private repositories; API keys only in gitignored .env files; strict contract validation; model artefacts verified by SHA-256 hashes	No authentication or HTTPS yet; required before deployment
Privacy	DMC recordings stored locally under numeric identifiers, used only for evaluation and deleted after use; external APIs can be switched off; offline student models avoid sending text externally	Met for research data; production data-protection controls pending
Usability and accessibility	Questions and speech in the caller's language; numbered choices when a sender is struggling; dashboard with light and dark themes that works at phone width	Formal operator usability study not yet conducted
Explainability	Class probabilities, MPDS evidence, merge evidence trails, review flags, SHAP analysis for urgency	Met
Maintainability	Versioned contract, provenance files and run identifiers for every model, immutable dataset versions, automated tests (28 test modules in C4)	Met
Scalability	Components are separate services that can be scaled independently	Single-process prototype; C3 keeps one global conversation; concurrent load not yet measured

4.4 Integrated workflow and exception handling

In normal operation, a call or SMS conversation runs to completion in C1 or C3, the report is published, C4 returns its first-stage results immediately, and the background worker decides and attaches the report to an incident within seconds. The dashboard picks up the change at its next poll. Table 4.4 lists the important failure paths and the behaviour designed for each.

Table 4.4: Failure paths and designed behaviour

Situation	Designed behaviour
Caller falls silent or the call drops	A silence event is recorded as a turn; after the idle timeout the call is finalised, C2 computes the verdict from the turns received, and the report is published with disconnected = true
SMS sender stops replying	The watchdog publishes a partial case; if the sender later completes it, the complete case supersedes the partial one
Sender replies with distress instead of an answer	The reply is not stored as a field value; a calming acknowledgement is sent and the question is repeated, in numbered form if fatigue is detected
LLM extraction invents or misquotes a value	The verbatim-quote check rejects the value; deterministic sources or a follow-up question are used instead
OpenAI, Google Places or the network is unavailable	C4 uses its offline mT5 student and cached place results; C3 falls back to deterministic templates; the LLM source is marked unavailable
C4 is slow or down	Publishing times out silently; the call or SMS conversation is unaffected
Malformed or tampered report	Rejected with HTTP 422 by the contract schema
Location cannot be resolved	The incident is flagged as unknown location and listed under the map; geography abstains instead of guessing
Two reports look alike but differ in a decisive detail	The evidence judge detects the contradiction (for example a different house number) and the reports stay separate
A report fits two open incidents	The decision is kept but the report is flagged for review as possibly linking the incidents

4.5 Team contribution and dependency matrix

Table 4.5 records the division of research, development and integration work. Each student owns one component end to end; integration and evaluation of the whole system were shared.

Table 4.5: Team contribution and dependency matrix

Member	Research and development	Integration work	Depends on / provides to
Edirisinghe A.K. (IT22129758)	Multilingual STT and Sinhala Whisper fine-tuning; language ID; XLM-R NER and relevance; readiness assessor; clarification policy; multilingual TTS; Twilio voice channel	Hosts C2 bridge; publishes calls to C4 via the shared client; panel console	Provides turns to C2 and reports to C4; uses C2 verdicts
Member	**Research and development**	**Integration work**	**Depends on / provides to**

Thanseel S.M. (IT22644480)	Simulated call corpus and labelling; DMC diarisation and evaluation; acoustic and text features; late-fusion urgency model; error and domain-shift analysis	urgency_bridge module; per-turn and call-level API; offline bridge tests	Uses C1 audio and transcripts; provides urgency to C1 and C4
Karunarathna D.M.S.G. (IT22263476)	SMS corpus; LLM cleaning chain and guards; noise taxonomy; adaptive router; hallucination benchmark; mT5 fine-tuning; distress, fatigue and MPDS priority	Publishes SMS cases and partial cases to C4; untruncated conversation log	Provides reports and MPDS codes to C4
Siriwardena A.N. (IT22504418)	SL-DisasterCoref benchmark; extraction student and reconciliation; evidence pool; geography and time models; LaBSE and mDeBERTa finetuning; fusion, decision and incident records; dashboard	c4.v2 contract and shared client v2; live integration check; launcher integration	Consumes C1, C2 and C3 outputs; provides the operator view
All members	Proposal, data analysis, ethics and MLOps reports; progress presentations	End-to-end testing of calls and SMS through C4; common report	—

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The components share Python 3 as the implementation language, Flask for services, Hugging Face Transformers and PyTorch for neural models [17], scikit-learn for classical models [18], and Git for version control. Table 5.1 lists the main technologies and the reason for each common choice.

Table 5.1: Common technology stack

Layer	Technologies	Reason for choice
Services	Flask, Flask-CORS, Pydantic	Lightweight HTTP services; strict schema validation for the shared contract
Telephony	Twilio Programmable Voice and SMS	One gateway for both channels; webhook model fits turn-by-turn dialogue
Speech	faster-whisper (CTranslate2, int8), Whisper-small fine-tuned for Sinhala, librosa, noisereduce, pyannote.audio	Multilingual STT that runs on CPU; acoustic feature extraction; diarisation for real calls
Language models	XLM-R, LaBSE, mDeBERTa-v3, mT5-small, multilingual NLI (MiniLMv2)	Pre-trained models that cover Sinhala and Tamil and can be fine-tuned on free GPUs
LLM services	OpenAI gpt-4o-mini and gpt-4o (optional)	Extraction, distress and severity checks where rules fail; teacher for students
Classical ML	scikit-learn, imbalanced-learn, SHAP	Interpretable models for small data; class balancing; explanations
Speech output	Piper (Sinhala), MMS-TTS, Edge-TTS	Language-matched voices for clarification
Geography and UI	Google Places (cached), GeoNames gazetteer, Leaflet with OpenStreetMap	Place resolution and the incident map
Development	Git and GitHub (private), VS Code, Google Colab (T4 GPU), pytest	Version control, GPU training and automated testing

Infrastructure was intentionally modest so that the prototype reflects what a public agency could operate. All inference runs on a CPU laptop; fine-tuning runs on Colab and each trained model is verified on the laptop (hash check, identical outputs, reproduced metrics) before installation. Data are stored as versioned files: JSONL

report stores, JSON incident stores, and immutable dataset versions, each built by a script from the previous version.

5.2 Development and integration process

Development followed an iterative process organised around the two progress presentations. After the proposal, each member built a baseline of their component on simulated or synthetic data. Feedback at the first progress presentation (PP1) was converted into measurable design changes: C1 replaced Whisper log-probability as a proxy for accuracy with word and character error rates against human transcripts and introduced a dialogue-aware clarification policy; C2 moved from early to late fusion, removed a noisy feature and adopted a better Sinhala ASR; C3 replaced a pure-LLM priority classifier, which missed messages such as “we can’t breathe”, with a deterministic protocol plus an LLM safety net; and C4 replaced static rule-based matching with a learned matching pipeline that separates location from semantic similarity.

Three practices were applied across the team to keep results honest. First, for every modelling step, baselines were compared with the proposed model, settings were chosen on development data, and test data were used once. Second, decision rules and thresholds were written down before results were seen, and negative results were recorded rather than discarded. Third, data and models were versioned: labels were frozen with SHA-256 hashes before experiments, and every installed model carries a provenance file with its training run identifier.

Integration work was interface-first, as described in Section 4.1. Changes to the shared contract were made together with the consuming code, verified with the live integration check, and followed by end-to-end conversation tests. End-to-end testing proved essential: it found defects that unit tests had not, such as a fatigue mode that reset between SMS turns and generic nouns such as “my house” accepted as locations.

Development was organised around the two progress-presentation milestones (PP1 and PP2) rather than fixed-length sprints, with a weekly check-in between all four members to review individual progress and keep the overall system design aligned, and markedly more frequent coordination in the two to three weeks immediately before

each milestone: near-daily Teams calls alongside in-person meetings. An MS Planner board was created for the project, but task-level tracking on it was not used consistently; day-to-day progress was instead coordinated through these recurring check-ins and each member's individual log book. Code review followed an ownerreview pattern rather than a formal pull-request rotation: each member developed their own component independently, and the team lead reviewed proposed changes before merging them into the shared integration branch. This covers the project's active development period, December 2025 to October 2026.

5.3 Deployment and operational configuration

The prototype is deployed as four local services opened from one launcher page. Table 5.2 lists the services and their configuration.

Table 5.2: Deployed services and configuration

Service	Address	Configuration
Launcher	localhost:5050	Component URLs set by environment variables
C1 call agent and panel console (with C2 in process)	localhost:5010; Twilio Voice webhooks via tunnel	.env: ports, model names, TTS engine, maximum turns; URGENCY_ROOT locates C2
C2 standalone web demo	localhost:5000	Same locked model and ASR pipeline as the live bridge
C3 SMS web chat and Twilio SMS webhook	localhost:5020 and :5021 via tunnel	.env: OpenAI key; template and keyword resources
C4 fusion API, /operator and /dashboard	localhost:8000	C4_LLM_ENABLED, C4_PLACES_LIVE, C4_5B_BACKGROUND, C4_EXTRACTOR; C4_URL, C4_TIMEOUT and C4_ENABLED for the shared client

Build and configuration are performed with pip requirement files and environment variables, so that behaviour changes without code changes. Every optional external dependency can be switched off: C4 runs without OpenAI (offline mT5 student) and with cache-only place lookup. Health is checked through the /health endpoint of C4 and the start-up warm-up of models in C1 and C2. Report and incident stores are append-only JSON files that survive restarts and can be backed up by copying; they

are excluded from version control because they contain personal data. For production, the team recommends a DMC-hosted server with a GPU, PostgreSQL storage with concurrent workers, HTTPS, role-based authentication, encrypted storage, audit logging and monitoring of latency and accuracy.

5.4 Evidence of team execution

Evidence of planned and coordinated execution is available from several sources. The private Git repositories record the development history of each component, including the integration commits for the shared client and the contract. The team delivered the proposal reports, the Data Analysis Report, the Ethical Analysis Report, the MLOps Report and two progress presentations, each prepared jointly, and the PP2 presentation documents how every PP1 comment was addressed by each member. Experiment records are kept as versioned outputs: the urgency component documents 55 reproducible analysis scripts behind its locked model, and the fusion component documents every step, the rule declared before each measurement and the result.

An exact count of Teams meetings was not logged; qualitatively, meetings were frequent throughout and especially dense in the weeks preceding each milestone, as described in Section 5.2. Supervisor meetings and research diary entries are evidenced individually in each member's log book (Appendix B). This author (Component 2) attended approximately 14 supervisor meetings and recorded approximately 74 dated diary entries between December 2025 and October 2026. Across the team, the individual log books record 77 diary entries for Component 1 (Edirisinghe, multilingual conversational AI), 74 for Component 2 (Thanseel, multimodal urgency detection), 95 for Component 3 (Karunarathna, SMS emergency intelligence) and 84 for Component 4 (Siriwardena, multi-source incident fusion and dashboard), totalling 330 logged diary entries across the team over the same period.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

The integrated evaluation answers five questions derived from the group objectives: (EQ1) does each component outperform a reasonable baseline on its own task; (EQ2) how well do the components generalise from simulated to real data; (EQ3) does the

system reduce the dangerous error in each task, namely under-triage and false merges; (EQ4) do the components operate together end to end, including under failures; and (EQ5) are decisions explainable to operators. Table 6.1 lists the datasets used and Table 6.2 the principal metrics and success criteria.

Table 6.1: Datasets used in the evaluation

Dataset	Size and content	Used by	Role
Simulated emergency calls (C1)	507 recordings in five language styles; 306 transcripts with 783 gold entity spans; 212 relevance-labelled utterances	C1	Training and held-out evaluation
Emergency ASR set	445 training and 49 validation audio–transcript pairs	C1	Whisper fine-tuning
Labelled urgency calls	267 calls (High 130 / Medium 82 / Low 55), kappa 0.77	C2	Training and 5-fold cross-validation
DMC real calls	59 calls from the 2025 floods and landslides (High 23 / Medium 14 / Low 22)	C2	External validation only
Emergency SMS corpus	864 messages in six language styles	C3	Development and benchmarks
C3 benchmarks	500-example hallucination benchmark; 26-case MPDS and paraphrase sets; 24case distress set	C3	Safety evaluation
SL-DisasterCoref	535 call and SMS reports in gold incidents (308 / 112 / 115)	C4	Training, DEV selection and single test
Live sessions	628 C1 session logs (276 final reports); live SMS through Twilio	All	End-to-end and regression testing

Table 6.2: Principal metrics and success criteria

Task	Metric	Baseline	Success criterion
C1 entity extraction and relevance	Entity F1; relevance F1	CRF, BiLSTM-CRF; LR, SVM	Fine-tuned model above all baselines
Task	**Metric**	**Baseline**	**Success criterion**
C1 clarification	Question diversity; safety pass rate; language match	Fixed template per field; English-only voice	Higher diversity at 100% safety; language match above baseline
C2 urgency	Accuracy, macro-F1, High-class recall	Early fusion; singlemodality models	Improvement on simulated and real views

C3 priority and safety	Exact MPDS match; dangerous under-triage; detector F1	Keyword-only priority; overlap and embedding detectors	Fewer under-triaged cases; best detector F1
C4 fusion	False merge, false split, link recall, merge precision, B^3 F1 [19]	Report-level matching; rules	False merge < 2%; merge precision > 90%

6.2 Integration and end to end testing

Table 6.3 summarises the testing carried out at integration and system level. Component-level experiments are reported in the individual theses.

Table 6.3: Integration and system tests

Test type	What was tested	Outcome
Unit tests	C4 adapter, extraction, evidence, geography, time, fusion, decision and revision (28 modules), including a test that urgency never enters fusion features; C3 agents	Pass
Interface tests	Offline C1–C2 bridge test and wiring dry run; contract validation with unexpected fields	Pass; drift rejected with HTTP 422
Live integration check	Real C4 server started; C1 and C3 sessions published through the real shared client, with and without the LLM step	Reports stored and attached to incidents
End-to-end conversations	Complete call and SMS conversations through all components, including distress replies, numbered answers and life-threat language	Two defects found and fixed (fatigue mode not persistent; generic nouns as locations)
Failure handling	Dropped calls (auto-commit 90 s, idle finalise 120 s); abandoned SMS; C4 outage; external APIs disabled	Reports published or conversations unaffected as designed
Regression	Full C1 pipeline re-run over 502 calls; scripted checks for triage routes, numerals and Sinhala / Tamil TTS	Pass
Consistency	C2 web demo against the live bridge	Byte-identical outputs
Live channel tests	Real calls through Twilio Voice and real SMS from a phone	Processed end to end

Two limits of the testing must be stated. Performance testing was limited to singleuser operation; concurrent multi-caller load has not been measured, and C3 currently keeps one global conversation. Security testing was limited to contract validation and secret

management, because authentication and transport encryption are not yet implemented.

6.3 Overall results

Table 6.4 brings together the principal results of the four components. Figures 6.1 to 6.4 show the most important comparison for each component and Figure 6.5 the operator dashboard.

Table 6.4: Principal results of the four components

Component	Measure	Proposed	Baseline
C1	Emergency NER, entity F1 (held-out test)	0.429 (fine-tuned XLM-R)	0.275 CRF; 0.278 BiLSTM-CRF
C1	Utterance relevance, F1	0.819 (fine-tuned XLM-R)	0.783 LR; 0.765 SVM
C1	Clarification question diversity (502 calls)	0.222, 100% safety pass	0.040 fixed template
C1	Calls rated not-ready-critical by readiness check	74.1%	91.4% null-field check
C1	Spoken language matches caller	100%	41% English-only voice
C2	Accuracy / macro-F1, simulated 5-fold CV (n = 267)	66.4% / 0.639	64.0% early fusion
C2	Accuracy / macro-F1, DMC scope-matched (n = 11)	80.0% / 0.753	—
C2	Accuracy / macro-F1, DMC full (n = 59)	44.8% / 0.431	34.2% early fusion; 45.8% text-only; 33.6% acoustic-only (accuracy)
C3	Hallucination detection, F1 (500 examples)	0.97 NLI entailment	0.92 overlap; 0.77 embedding
C3	MPDS exact match / dangerous under-triage (26 paraphrased)	73.1% / 1 case	57.7% / 11 cases
C3	Distress detection F1 (24 cases)	1.00	0.92 first prompt
C3	LLM calls avoided by adaptive router	58.3% (2,592 → 1,080)	All stages for all messages
C4	Field extraction, mean correct (DEV / TEST)	0.788 / 0.73 (mT5 student)	0.155 / 0.12 upstream
Component	**Measure**	**Proposed**	**Baseline**

C4	Same-incident AUC, semantic channel (TEST)	0.946 fine-tuned LaBSE	0.846 off-the-shelf; 0.621 character TFIDF
C4	False merge / merge precision (TEST)	0.9% / 98.1%	Target < 2% / > 90%
C4	False split / link recall / B^3 F1 (TEST)	28.0% / 70.7% / 0.864	14.7% / 84.0% / 0.921 without safety rules

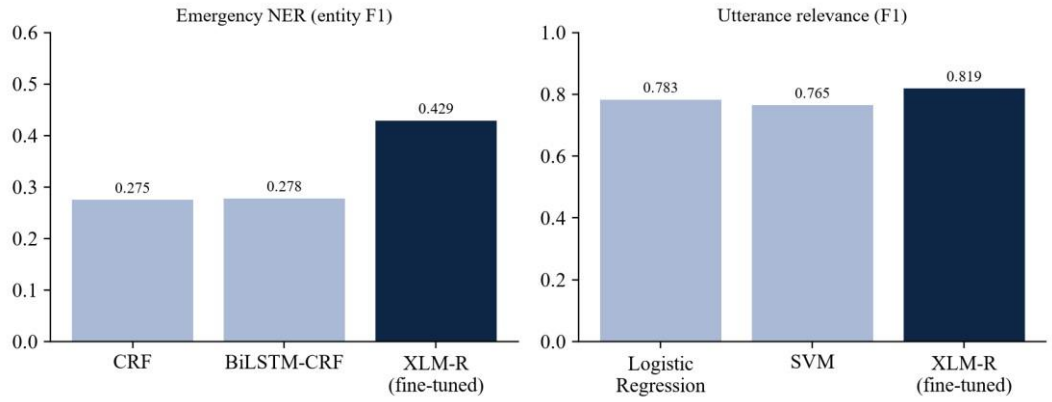


Figure 6.1: C1 model comparisons for emergency entity extraction and utterance relevance

Component 1. The fine-tuned XLM-R models outperform the classical baselines for both entity extraction and relevance (Figure 6.1). Entity F1 remains modest at 0.429 and varies by language (0.577 for Singlish, 0.471 for Sinhala, 0.407 for English, 0.235 for Tamil and 0.20 for Tanglish), which reflects the small labelled corpus. The adaptive policy produced five times more varied questions than a fixed template while passing every safety check, and the readiness assessor reduced the share of calls marked as critically incomplete from 91.4% to 74.1% by recognising information that a null-field check misses.

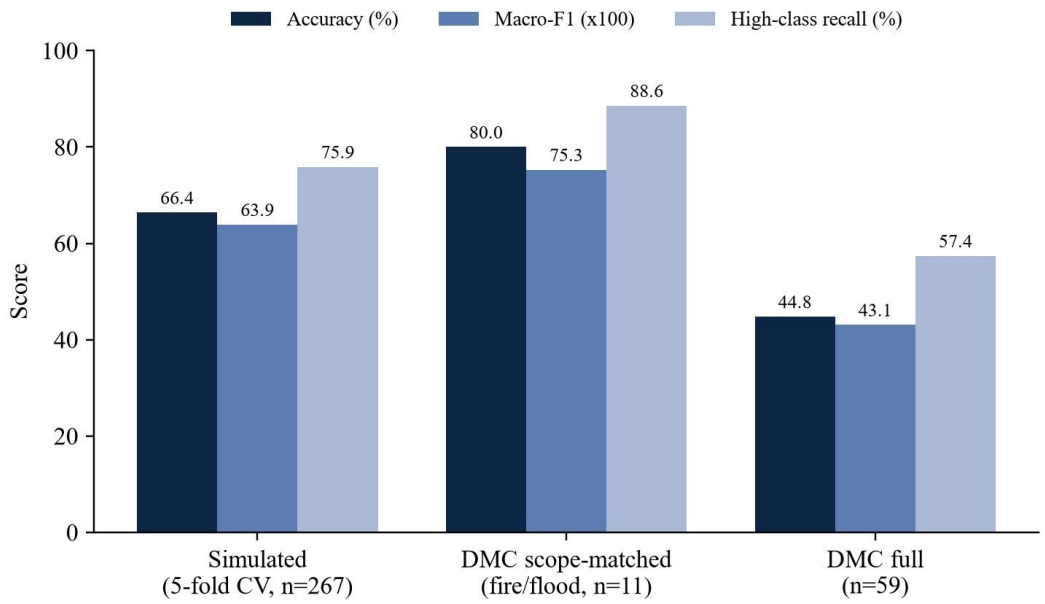


Figure 6.2: C2 urgency detection across the three evaluation views

Component 2. The late-fusion model improves on early fusion on every view (Figure 6.2). Its High-class recall, the measure that matters most for safety, is 75.9% on simulated calls, 88.6% on scope-matched DMC calls and 57.4% on all DMC calls. The drop from simulated to full real data is explained largely by emergencytype coverage: on the fire and flood calls that the simulated corpus was built around, accuracy recovers to 80.0%. Speech recognition on real telephony audio is the dominant remaining source of error.

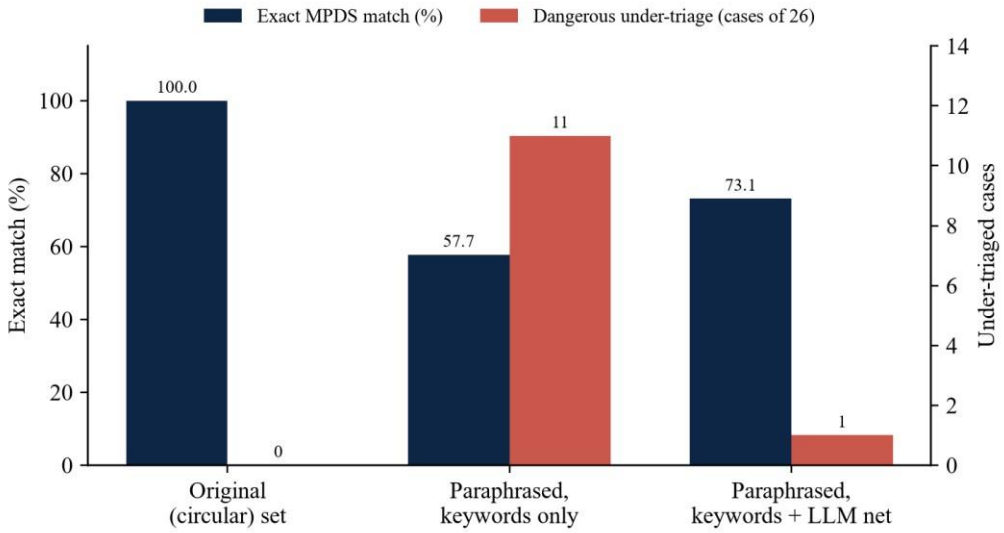


Figure 6.3: C3 MPDS priority accuracy and dangerous under-triage on 26 scenarios

Component 3. The first MPDS benchmark scored 100%, but because its examples used the same vocabulary as the keyword lists, it was replaced with a paraphrased version that avoids every keyword. Keyword-only priority then fell to 57.7% and under-triaged 11 of 26 life-threatening cases. Adding an LLM severity check that can only raise priority restored 73.1% exact match and left one under-triaged case (Figure 6.3). A cross-model comparison also showed that gpt-4o-mini agreed with gpt-4o on only 4.5% of non-null location extractions, because the smaller model labels generic nouns as places; a guard now rejects such values.

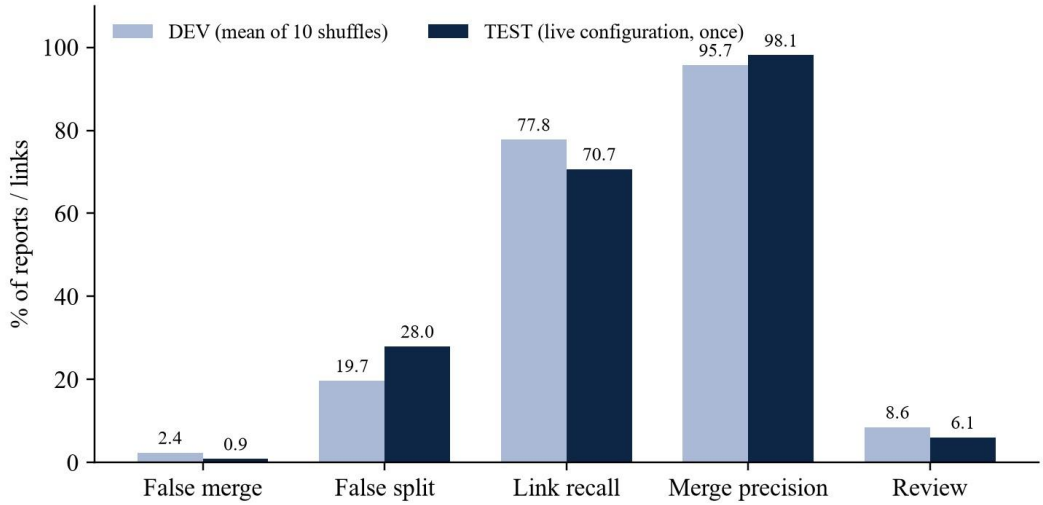


Figure 6.4: C4 incident-fusion decision outcomes on development and test reports

Component 4. With the live configuration, C4 made false merges for 0.9% of test reports with 98.1% merge precision, meeting the safety targets (Figure 6.4). The price is a false-split rate of 28.0% on test: about one in four true links is missed, most often between reports in different scripts. This is the less dangerous error by the cost reasoning behind the design, since a missed link leaves a duplicate record rather than leaving a household without help. Without the researcher's safety rules, false splits fall to 14.7% at the cost of more false merges (1.7%), which quantifies the trade-off for the DMC to set.

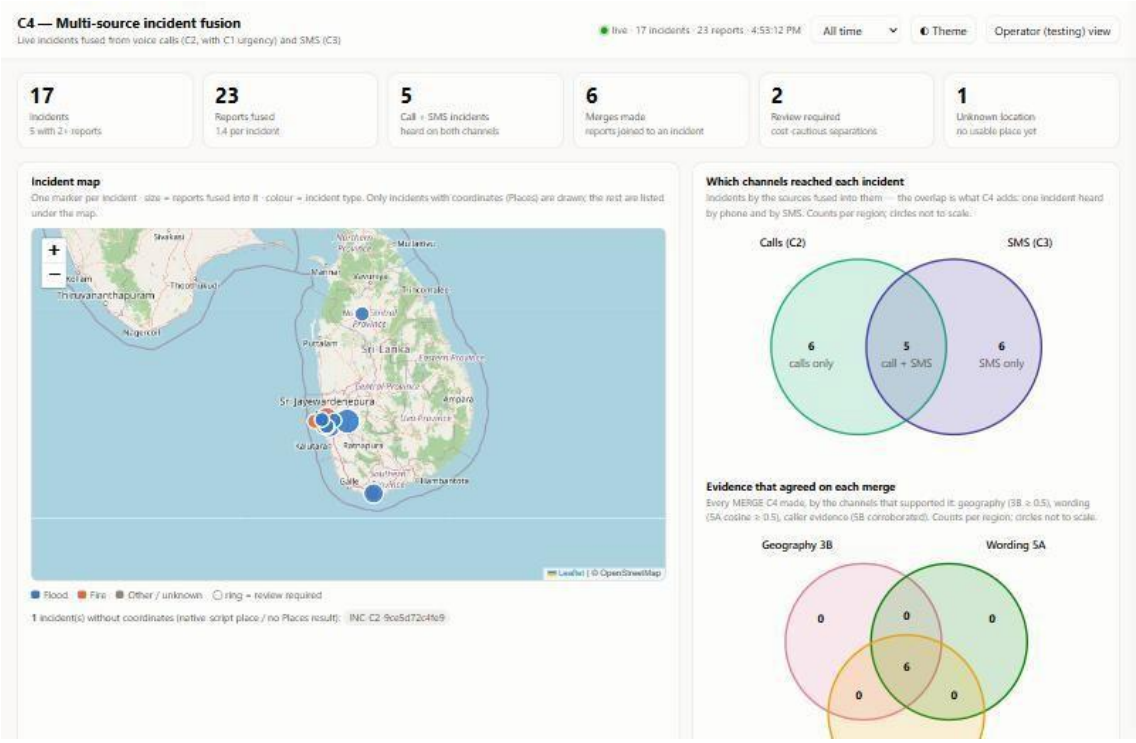


Figure 6.5: Operator dashboard showing fused incidents from calls and SMS (original in colour)

Figure 6.5 shows the operator dashboard during a live session, with 17 incidents fused from 23 call and SMS reports, five incidents heard on both channels, and two decisions flagged for review. (The dashboard labels use the internal identifiers of the prototype for the call and SMS sources.) Each incident opens a drawer with its fused record, timeline, evidence trail and the priority of every member report.

6.4 Stakeholder and user validation

Validation with stakeholders took three forms. First, external validation on real data: the DMC's 59 usable recordings were annotated before any model was applied and used to test C2 and to diagnose speech recognition failures, which showed that the simulated-to-real gap is driven by telephony audio quality and by emergency types missing from the simulated corpus. Second, native-speaker review: Sinhala and Tamil question templates in C3 were reviewed and corrected by native speakers (for example, a machine-translated Sinhala location question was replaced with a more natural phrasing), and verification worksheets were used to check diarised DMC caller turns. Third, expert review by the supervisors and the progress review panels, whose comments led to the changes described in Section 5.2.

A structured usability evaluation of the complete system with DMC operators, such as a simulated drill scored with the System Usability Scale [20], has not yet been carried out. This is the most important missing piece of validation evidence, and it is listed first among the future directions in Chapter 8. Remaining stakeholder concerns are the speed of speech recognition on CPU hardware, the coverage of Tamilspeaking callers in real data, and the review workload that the cautious fusion setting creates.

No DMC or operator feedback session had been conducted as of the time of writing. One is planned for 15 October 2026; its date, participants and findings will be added once it has taken place. This remains the most important outstanding piece of stakeholder validation for the integrated system (Section 6.6, Section 8.3).

6.5 Achievement of common objectives

Table 6.5 maps the evidence to each group objective.

Table 6.5: Achievement of the group objectives

ID	Status	Justification
GO1	Achieved	Live calls and SMS in Sinhala, Tamil, English and code-mixed forms were processed end to end into incidents; per-language results are reported, with weaker results for Tamil and Tanglish
GO2	Achieved	Adaptive clarification on voice and SMS; readiness assessment; distress and fatigue handling validated on benchmarks
GO3	Partly achieved	Every report carries an urgency or MPDS signal and the dangerous error was reduced, but urgency accuracy on all real DMC calls (44.8%) is not yet adequate for operational reliance
GO4	Partly achieved	False merge 0.9% and merge precision 98.1% meet the targets; link recall (70.7%) leaves many duplicates unmerged
GO5	Achieved	Probabilities, MPDS evidence, merge evidence trails and review flags are shown; the operator makes every decision
GO6	Achieved for the prototype	Versioned contract, live integration check and failure tests pass; concurrent load and production hardening remain

6.6 Integrated limitations and threats

- **Speech recognition on real telephony audio** is the most important system-level limitation. Because C2's text features and C4's evidence both depend on C1's transcripts, recognition errors propagate through every later stage.

- **Simulated and synthetic development data.** Most development data were scripted or synthetic. Actor speech differs from real panic, and synthetic SMS cannot reproduce real network corruption. Real-data results are available only for C2 and on a small sample (59 calls, 11 scope-matched).
- **Language imbalance in real data.** 58 of the 59 DMC calls are in Sinhala, so performance for Tamil-speaking callers on real calls is largely untested; place resolution is also weaker for Sinhala and Tamil script (53–58% within 1 km) than for Latin script (83%).
- **Emergency-type and regional coverage.** The urgency corpus centres on fire and flood, and the DMC calls come from a limited set of districts.
- **Dependency between components.** C4's decisions depend on the quality and completeness of C1 and C3 extraction; an unresolved location or a wrong incident type upstream reduces fusion quality.
- **Latency and concurrency.** CPU speech recognition is too slow for live calls at scale, and concurrent operation has not been measured.
- **Reuse of test data.** Some C4 test results were measured more than once as the system evolved; this is documented in the individual thesis but weakens their independence.
- **External services and licences.** Optional dependence on OpenAI and Google Places raises availability, cost and data-protection questions, and operational use of MPDS requires a licence from its owner.

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The integrated product offers disaster-response agencies one triage layer for both voice and SMS that understands Sri Lankan languages, asks the questions that matter, ranks reports by urgency and consolidates duplicates into explainable incident records. Its value is created for several groups. Callers benefit from being understood in their own language, from shorter and simpler questioning, and from an SMS path when calls fail. Call-takers receive more complete reports and an urgency cue for every call. Dispatchers and EOC supervisors see one record per incident with its evidence, instead

of many duplicates, which supports better allocation of scarce responders. Society benefits from more equitable access to emergency services for Tamil-speaking and code-mixing communities, and from a reusable foundation of Sinhala and Tamil emergency language resources.

7.2 Market alternatives and differentiation

The primary target segment is national disaster-management and emergencyresponse agencies in Sri Lanka, beginning with the DMC and extending to provincial emergency operations, police emergency services and municipal emergency units. A secondary segment is agencies in South Asia that face similar multilingual, lowresource conditions. Table 7.1 compares the integrated product with the alternatives available to these customers.

Table 7.1: Alternatives and differentiation

Alternative	Strengths	Limitations for the target segment
Manual call-taking and spreadsheets	No technology cost; full human judgement	Does not scale during surges; no deduplication; urgency depends on operator load
Commercial CAD systems	Mature dispatch workflows and integrations	English-centric; one record per call; high licence cost; no Sinhala / Tamil language support
Crisis social-media tools (e.g. AIDR)	Large-scale classification of public posts	Do not handle private calls or SMS; English-focused; no structured dispatch output
General-purpose	Fluent multilingual text	Hallucination risk; no urgency from
Alternative	**Strengths**	**Limitations for the target segment**
chatbots and LLM APIs		voice; no incident fusion; external data transfer
This integrated system	Sinhala, Tamil and codemixed support; voice and SMS; prosodic urgency; guarded extraction; crosslingual incident fusion; explainable decisions	Prototype maturity; real-data validation still limited

The differentiation is defensible because it rests on assets that are costly to reproduce: emergency-domain datasets and benchmarks in Sinhala and Tamil (including the SL-DisasterCoref fusion benchmark and the labelled urgency corpus), models fine-tuned

on them, and an architecture designed around the asymmetric costs of emergency errors.

7.3 Business model and feasibility

The proposed model follows the government-technology licensing pattern that suits public-sector procurement in Sri Lanka, introduced in stages as evidence accumulates (Table 7.2). The modular architecture allows the SMS channel or the fusion dashboard to be adopted on its own by agencies that are not ready for the full system.

Table 7.2: Staged business model

Stage	Offering	Revenue or funding
1. Research pilot	Supervised pilot with the DMC on recorded and live drills	Research grants (for example NSF Sri Lanka) and sponsored research agreement
2. National deployment	On-premises licence for the DMC with hosting support, maintenance and operator training	Annual licence and one-time implementation fee
3. Modular expansion	SMS triage or incident dashboard for provincial units, police and municipal emergency services	Per-site or per-district subscription
4. Regional adaptation	Adapted language models and gazetteers for other South Asian agencies	Regional licence and customisation services

The research phase was planned within a one-year project budget of LKR 120,000 covering software licences, cloud services, commercialisation and IP activities, and miscellaneous costs. Operating costs are dominated by a GPU server for speech recognition, telephony charges and optional LLM usage; the offline student models and the adaptive router (58.3% fewer LLM calls) were built specifically to reduce recurring API cost. Technical feasibility is shown by the working end-to-end prototype on commodity hardware; operational feasibility will require the GPU deployment, concurrency testing and operator validation described in Chapter 8. Indicative licence prices estimated at the proposal stage must be validated with the DMC before they are relied on.

7.4 Adoption intellectual property and compliance

Adoption barriers. Public agencies adopt life-safety systems cautiously. The main barriers are trust in automated decisions, the need for on-premises hosting, integration with existing EOC workflows, operator training and procurement timelines. The design responds by keeping the operator in control, showing evidence for every decision and allowing staged, modular adoption.

Intellectual property. The team's IP comprises the integrated architecture, the component software, the datasets and benchmarks, and the fine-tuned models, subject to SLIIT's intellectual property policy and any agreement with the DMC. Registration of software copyright with the National Intellectual Property Office of Sri Lanka and publication of research results are planned. Third-party models are used under their licences (for example, Whisper and XLM-R under MIT and LaBSE and mT5 under Apache 2.0), which permit commercial use with attribution. MPDS is a proprietary protocol; the prototype uses only its conceptual priority structure, and operational use of the protocol would require a licence. Google Places results are subject to the provider's terms on caching and display, which must be reviewed for production.

Data protection and ethics. Emergency calls and messages contain personal and sensitive data, including voices, phone numbers, addresses and health information. Production use must comply with Sri Lanka's Personal Data Protection Act No. 9 of 2022 [21], including a lawful basis for processing, purpose limitation, retention limits and security safeguards. During the research, DMC recordings were used only with the DMC's permission, stored locally under numeric identifiers, never uploaded to external services, used only for evaluation and scheduled for deletion after use, and only aggregate results and short anonymised fragments are reported. Sending report text to external APIs can be disabled.

Safety and professional responsibility. The system is decision support, not an autonomous dispatcher. Its known failure modes (under-triage on real calls, missed cross-lingual links, weaker performance for Tamil) are documented so that the DMC can decide where human handling must remain the default. Bias is monitored by reporting results per language and per class.

7.5 Sustainable Development Goal contribution

The plausible contribution to SDG targets 11.5 and 13.1 is faster and more complete understanding of emergency reports during climate-related disasters, which is a precondition for timely response. The contribution to target 3.d is earlier recognition of life-threatening situations, and the contribution to target 10.2 is service in the languages that people actually use. These contributions are, at present, potential: they have been demonstrated on prototype data and have not yet been measured as outcomes in operation.

Possible negative effects must also be acknowledged: over-reliance on automated urgency could delay help for under-triaged callers; unequal performance across languages could disadvantage Tamil speakers; and storing emergency data creates privacy risks. To sustain benefits and limit harms, the team recommends keeping human review of every dispatch, reporting performance by language before and during deployment, collecting and labelling more real Tamil and non-flood calls, retraining with operator-confirmed incidents under drift monitoring, and building local capacity at the DMC to operate and audit the system.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The project set out to determine whether an AI-supported triage system could clarify, prioritise and fuse multilingual emergency calls and SMS messages in Sri Lankan low-resource languages while keeping every decision explainable and under human control. The integrated evidence supports a qualified yes. A working system now takes calls and messages in Sinhala, Tamil, English and code-mixed forms, asks targeted questions in the caller's language, attaches an urgency or dispatch priority to every report, and consolidates reports across channels and scripts into incident records that operators can inspect. Each component improves on its baseline, and the design reduces the dangerous error in each task: dangerous SMS under-triage fell from 11 to 1 of 26 cases, and false merges of separate incidents were held below 1% on held-out reports.

The evidence also shows where the solution is not yet ready. Urgency accuracy on the full set of real DMC calls (44.8%) and the rate of missed incident links (28% false

splits) are too weak for operational reliance, and both trace back in part to a shared cause: the quality of speech recognition and extraction on real, noisy, crosslingual input. The system is therefore best understood as a validated research prototype that demonstrates feasibility and identifies precisely what must improve before deployment.

8.2 Principal integrated contribution

Beyond the four individual components, the team's integrated contribution is fourfold. First, the first end-to-end triage pipeline for Sinhala and Tamil emergency communication that joins voice clarification, prosodic urgency, SMS triage and cross-channel incident fusion. Second, a versioned integration contract that carries the conversation itself, not a summary, so that downstream components reason over the caller's own words with question–answer context. Third, a shared safety design across components: verbatim grounding of extracted values, asymmetric treatment of dangerous errors, urgency kept out of merge decisions, and review flags instead of silent automation. Fourth, reusable resources and evidence for low-resource emergency language technology: labelled call, SMS and fusion datasets, fine-tuned models, a real-call evaluation and a documented record of negative results.

8.3 Future directions

The future directions are listed in order of priority in Table 8.1.

Table 8.1: Prioritised future directions

Priority	Direction	Purpose
1	Operator validation: simulated drills with DMC call-takers and dispatchers, usability (SUS) and workload measurement	Provide the missing stakeholder validation of the complete system
2	Telephony-domain speech recognition: fine-tune on real 8 kHz call audio and move to GPU or streaming STT	Address the dominant shared error source and the latency limit
3	More real and diverse data: Tamil calls, non-fire / flood emergencies, more districts, native review of keyword lists	Reduce language and scenario bias; strengthen real-data results
4	Cross-lingual linking: more cross-script sameincident training data and better place resolution for Sinhala and Tamil script	Reduce false splits without raising false merges

5	Production hardening: database storage, concurrent workers and per-sender memory, HTTPS, authentication, audit logs, load testing	Prepare for multi-user operation on DMC infrastructure
6	Feedback loop: retrain from operator-confirmed incidents with drift monitoring and experiment tracking	Sustain accuracy as real data accumulate

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APPENDICES

Appendix A Integrated interface specification

This appendix specifies the integration interface published by C4 and consumed by C1, C3 and the dashboard. The schema is defined in component4/service/contract.py and the client in shared/c4_client.py.

Table A.1: C4 REST endpoints

Method and path	Purpose
POST /v2/reports	Submit a report in the c4.v2 contract; returns the first-stage result or HTTP 422
GET /v2/reports, GET /v2/reports/<id>	List stored reports or fetch one report with all stage outputs
GET /v2/incidents, GET /v2/incidents/<id>	List incidents or fetch one fused incident record
GET /health	Service health check
GET /operator, GET /dashboard	Operator testing view and incident dashboard
GET /demo, POST /v2/test/reset	Turn-by-turn demonstration page and reset of its separate test store

An illustrative c4.v2 payload for an SMS case published by C3 is shown below (values shortened).

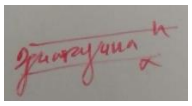
```
{
  "contract_version": "c4.v2",
  "report_id": "sms-20261003-0007",
  "source_component": "C3",
  "channel": "sms",
  "language": "singlish",
  "turns": [
    {"speaker": "caller", "text": "14 pansala para gampaha watura gaalanawa help"},
    {"speaker": "system", "text": "How many people are with you?"},
    {"speaker": "caller", "text": "api 4 denai"}
  ],
  "upstream": {"Incident": "flood", "Location": "14 pansala para gampaha",
               "People affected": "4", "Help needed": "boat rescue"},
  "priority_code": "CHARLIE",
  "disconnected": false,
"case_status": "complete"
}
```

Note: the source_component values in the code (“C2” for the call agent and “C3” for the SMS service) are internal identifiers of the prototype and predate the component numbering used in this report.

Appendix B Team contribution declaration

The following record of individual and shared contributions was agreed by all four members.

Table B.1: Team contribution declaration

Member	Individual contribution	Shared contribution	Signature
Edirisinghe A.K. (IT22129758)	Component 1: multilingual conversational AI for rapid clarification	Integration with C2 and C4; end-to-end testing; group reports	
Thanseel S.M. (IT22644480)	Component 2: multimodal urgency detection; DMC data preparation and evaluation	Urgency bridge integration; group reports	
Karunarathna D.M.S.G. (IT22263476)	Component 3: low-bandwidth SMS emergency intelligence	SMS publishing to C4; end-to-end testing; group reports	
Siriwardena A.N. (IT22504418)	Component 4: multi-source incident fusion, unified incident records and operator dashboard	c4.v2 contract and shared client; live integration check; group reports	